K. M. Abesh Ahsan

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Education

Islamic University of Technology (IUT) — Jun 2021 – October 2025
Bachelor of Science in Computer Science and Engineering (CSE) — Gazipur, Bangladesh
CGPA: 3.72/4.0

Research

Weakly Supervised Semantic Segmentation With Image Labels — January 2024 – October 2025

I designed a transformer-based approach to enhance object detail recognition in weakly supervised segmentation. Using UniCL with a Swin Transformer, I improved class activation map (CAM) quality and fine-grained feature extraction via windowed attention. My method incorporated advanced CAM refinement, affinity calculations, and encoder intermediate features. This model generated highly detailed masks for most classes, though accuracy dropped for a few categories like person and chair. Achieving a mean IoU of 50%, the approach delivered substantial fine-grained segmentation improvements, despite being below the state-of-the-art result of 74%.

Experience

Battery Low Interactive – Game Development Trainee — Oct 2024

During my industrial training as a Game Developer Trainee, I learned to conceptualize and program games from inception using Unity and C#. I engineered several mini-games to grasp mechanics, scoring algorithms, and physics-driven interactions. For my final project, I developed a 2D centrifugal-force game in which players maneuvered a stone to strike targets, implementing custom game logic, score tracking, and responsive UI components. I also integrated animations, curated assets, and designed scenes. I also explored the fundamentals of AR/VR during that time.

Skills

Programming Languages: Python, C++, JavaScript, TypeScript, Dart, Java, C#, C, SQL

Frameworks & Libraries: PyTorch, YOLOv5, Transformers (Hugging Face), OpenCV, NumPy, Matplotlib, PIL, React, Next.js, Node.js, Flutter, FastAPI, Tailwind CSS

Tools & Technologies: Git, Docker, MongoDB, MySQL, Redis, Jupyter

Projects

Semantic Segmentation From Scratch — **2024** — PyTorch, NumPy, Matplotlib, PIL

- Implemented U-Net and DeepLabV3+ models from scratch in PyTorch
- Trained on Carvana (binary) and Cityscapes (multi-class) datasets with visualization tools
- Evaluated using pixel accuracy and mIoU for autonomous driving applications

Sign Tutor AI — **2025** — Python, PyQt6, YOLOv5, OpenCV, PyTorch, NumPy

- AI-powered sign language learning app with real-time YOLOv5 gesture detection
- Gamified learning experience with instant feedback on user performance
- PyQt6 desktop interface with camera-based gesture recognition

End to End Semantic Segmentation — **2026** — Pytorch, PIL, Docker, FastAPI, Hugging Face, React

- Complete ML pipeline from data preparation to real-time inference and deployment
- Modular Python backend with Docker containerization for reliable serving
- REST API with FastAPI and interactive React frontend for image upload and visualization

PlayClassix (*Live demo*) — **2026** — Next.js, TypeScript, Tailwind, Zustand, Pusher, Upstash Redis

- Browser-based gaming platform with mini-games (Memory Match, Chess, Wordle)
- Real-time multiplayer via WebSockets with anonymous no-login entry
- Session-persistent user IDs and responsive cross-device design
- Serverless Redis storage

Photo Wizard — **2024** — NumPy, PyQt6

- Desktop image editor with 13+ features (crop, resize, blur, sharpen, filters, etc.)
- Custom filter creation support with all processing **from scratch** using NumPy

Programming Acheivements

IUT Intra First-Year Programming Contest 2021 — **2021** — 1st out of 32 teams

CodeRush 1.0: Programming Competition — **2023** — 7th out of 37 teams